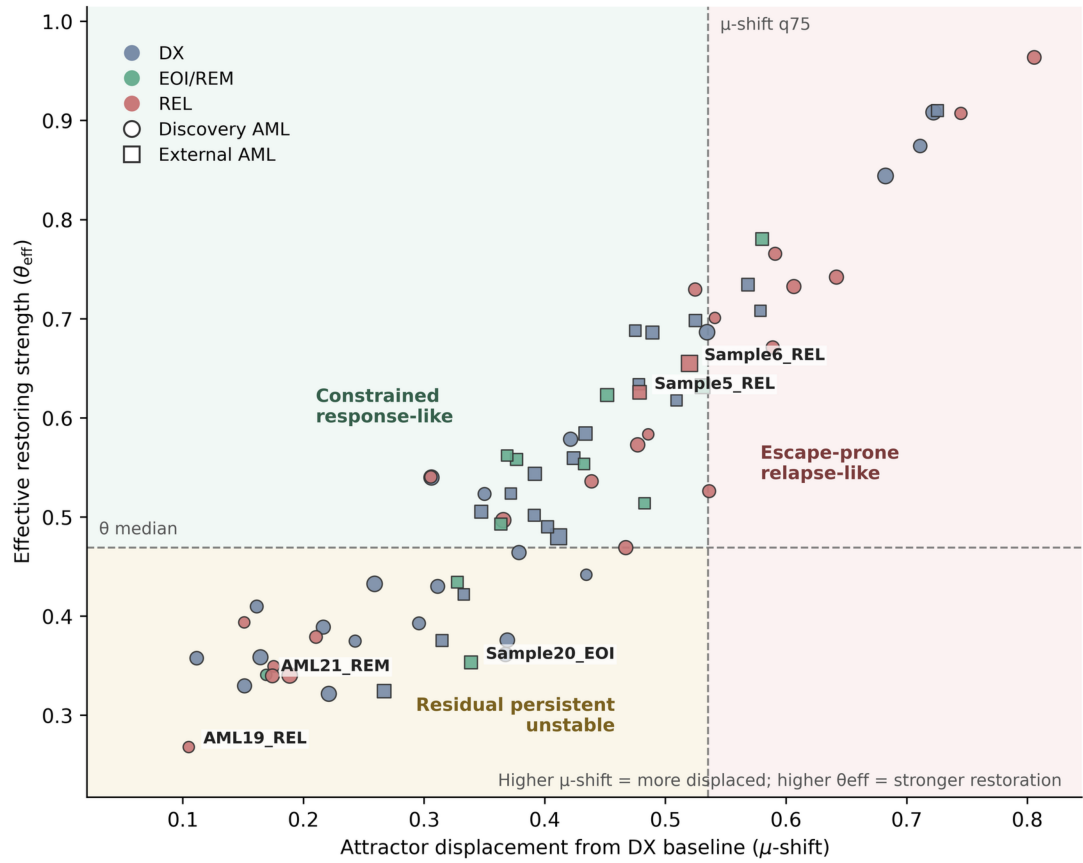


A Dynamic clinical map

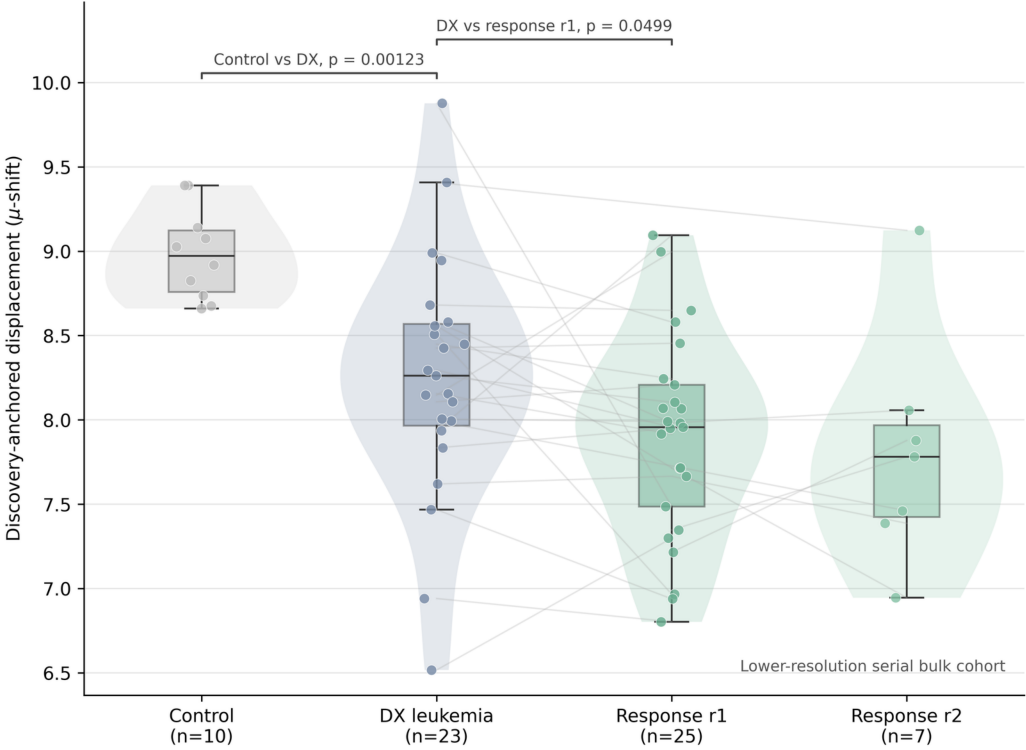


B Representative dynamic-state scorecard

Representative sample	θ_{eff}	σ_{eff}	μ -shift	Branch	State tier	Jump / switch
AML22_DX Discovery DX DX	0.686	0.587	0.535	Progenitor	Constrained	-0.54 / 0
AML21_REM Discovery EOI/Residual EOI/REM	0.341	0.981	0.169	HSC	Residual	—
AML16_REL Discovery REL REL	0.526	0.782	0.536	HSC	Escape-prone	0.67 / 1
Sample9_DX External AML DX DX	0.686	0.645	0.489	Mono/DC	Constrained	—
Sample20_EOI External AML EOI EOI/REM	0.353	0.948	0.339	HSC	Residual	—

Metric colors show column-wise standardized relative values; numbers show raw values.

C Serial bulk validation preserves disease-state ordering



D Clinical translation summary

Discovery cohort (GSE235063)

21 DX→REL intervals support both branch-continuous and switching routes.

Branch-continuous = 9
Branch-switching = 12

Top escape context:
GMP branch
mean jump = 1.58

External AML calibration (GSE235923)

An independent AML cohort reproduces the treatment-aware dynamic logic.

Median μ -shift
DX = 0.424
EOI/REM = 0.405
REL = 0.499

Full triads = 2

Serial bulk validation (GSE163634)

A lower-resolution, more deployable assay still preserves directional state structure.

Median μ -shift
Control = 8.972
DX = 8.263
 $r1 = 7.957$
 $r2 = 7.781$

Clinical translation takeaway

Dynamic states support response monitoring and residual-state triage.

Constrained: 22
Residual: 37
Escape-prone: 15

Examples:
Residual → AML21_REM, Sample20_EOI
Escape → AML16_REL

Together, these results position OU-Lévy-Branching as a clinically interpretable framework for dynamic response monitoring, residual-state interpretation, and relapse-risk assessment in pediatric leukemia.